

COMPUTER NETWORKS CMPG 325
INDIVIDUAL SEMESTER PROJECT
CLIENT DESIGN REVIEW

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1. Executive Summary

Student	SEDZE, TK
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Project ID	CMPG325-2026-121
Client ID	CLI-121
Assigned Organisation	Platinum Sound Recording Studio (Rustenburg)
Industry	Entertainment

Platinum Sound Recording Studio (Rustenburg, Entertainment industry) requires a segmented, secure network supporting day-to-day studio/admin/production operations, centralised file storage, secure remote device management, and limited after-hours wireless access for cleaning/security contractors. The proposed design uses VLAN segmentation, L3 core, VLSM addressing on the assigned 192.168.52.0/24 block, and SSH-only device management.

2. Assumptions

Assumption	Justification
Studio operates with 3 functional zones: Admin, Recording Studios (A & B combined into one VLAN), Production/Editing	Typical structure for a small-to-medium recording studio. Reception/business admin is separate from creative production spaces
15 - 25 devices per zone, fits comfortably in /27 subnets = 30 usable hosts	A boutique studio, not an enterprise; VLSM sized to realistic scale with room to grow
One shared audio/file server for session files	Recording studios routinely centralise project files for backup and cross-studio access
Contractors bring their own devices such as phones/tablets needing internet-only access, not internal resources	This matches "limited wireless access" in CR14 precisely
No VoIP/phone system requirement stated, so none was designed	This avoids inventing scope beyond the brief and sticks to what we were given

3. Client Requirements Analysis

Functional requirements:

- Wired connectivity for Admin, Studios, Production with logical separation to contain broadcast domains and simplify troubleshooting per zone
- Centralised server access from relevant zones (Admin + Production, primarily)
- Secure, auditable device management. There are no cleartext protocols on infrastructure devices

Non-functional requirements:

- Scalability: addressing plan leaves ~124 addresses unused in the /24 for growth
- Security: management plane isolated to SSH from a defined management subnet only; guest wireless isolated from internal VLANs
- Maintainability: clear VLAN-per-function mapping, consistent naming, documented in GitHub

4. Design Constraint and Change Request (CR-14)

Constraint/CR14: "After-hours cleaning and security contractors need limited wireless access."

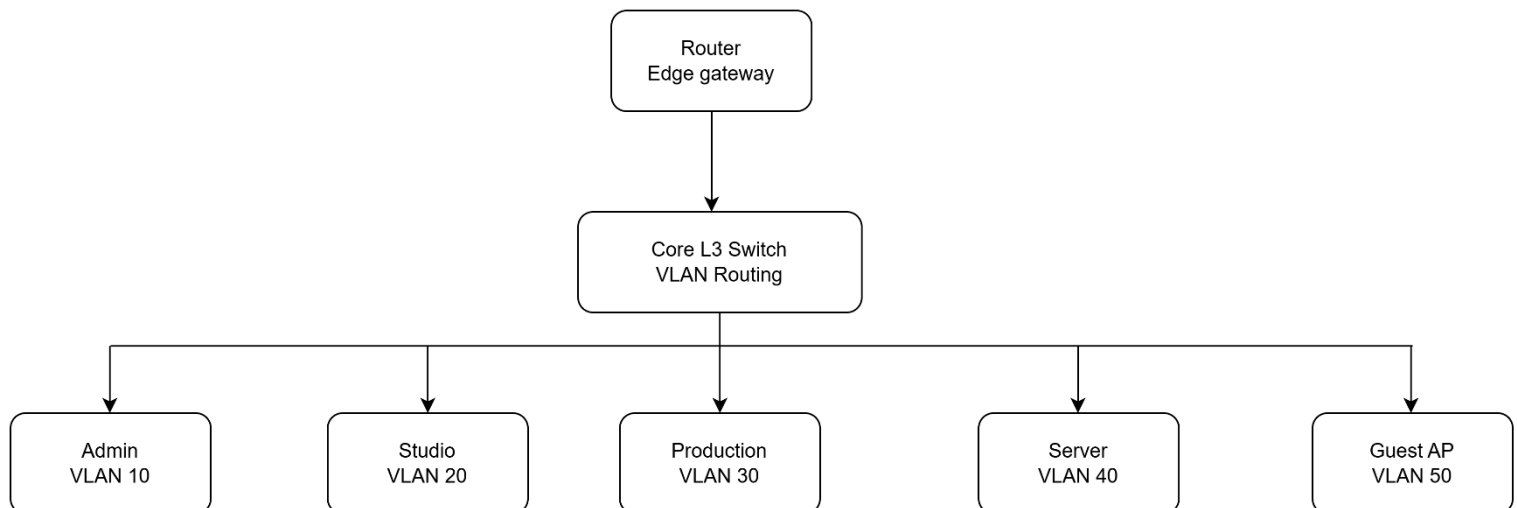
Design response:

- Dedicated VLAN 50, subnet 192.168.52.112/28, physically served by a wireless AP separate from any internal SSID
- ACL on the core device: permit DHCP/DNS/HTTP(S) outbound to internet, explicitly deny any traffic sourced from VLAN 50 destined for VLANs 10/20/30/40
- Time-based restriction using an IOS time-range object e.g. contractor-hours, 18:00 - 06:00 daily applied to the permit statement governing VLAN 50's internet access. This operationalises "after-hours" literally rather than just labelling the SSID "guest."
- Contractors are transient, non-employee users; isolating them protects client session files/audio IP and satisfies a reasonable security posture for a creative business handling client-owned content.

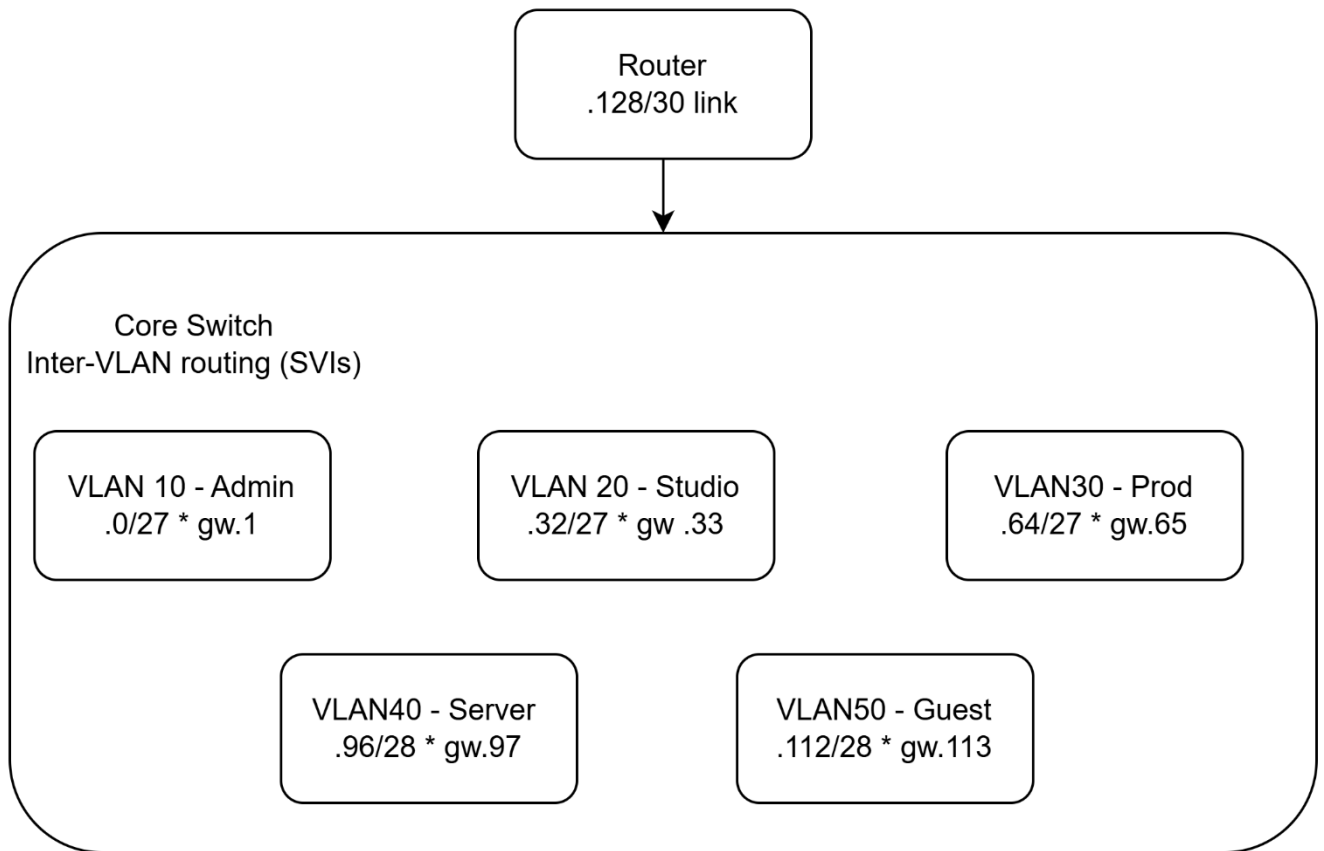
5. Topology and design rationale

- **Core L3 switch performing inter-VLAN routing** was chosen over a router-on-a-stick because switch backplane routing is faster for inter-VLAN traffic e.g. Production pulling files from the Server VLAN than trunking everything back through a single router interface which is relevant even at this scale.
- Router is retained solely as the edge/WAN gateway, keeping its role minimal and its attack surface small.
- Access switches per department keep failure domains contained, if the Studio switch fails, Admin and Production stay up.

Physical Topology



Logical topology (VLANs and addressing)

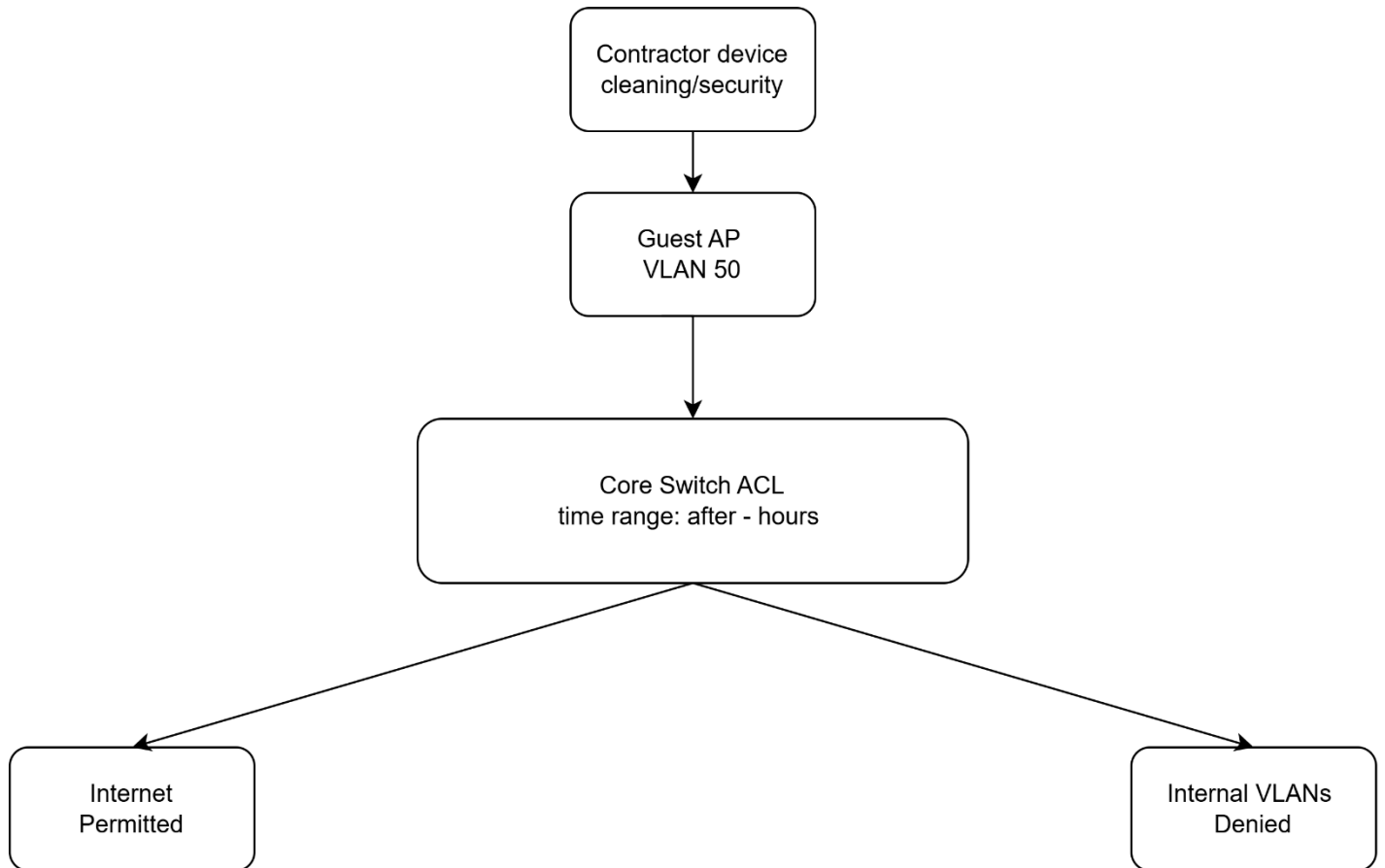


6. IP Addressing Plan

VLAN	Purpose	Subnet	Mask	Usable range	Gateway	Broadcast
10	Admin/Management	192.168.52.0/27	255.255.255.224	.1 – .30	192.168.52.1	192.168.52.31
20	Recording Studios	192.168.52.32/27	255.255.255.224	.33 – .62	192.168.52.33	192.168.52.63
30	Production/Editing	192.168.52.64/27	255.255.255.224	.65 – .94	192.168.52.65	192.168.52.95
40	Servers	192.168.52.96/28	255.255.255.240	.97 – .110	192.168.52.97	192.168.52.111
50	Guest wireless (CR14)	192.168.52.112/28	255.255.255.240	.113 – .126	192.168.52.113	192.168.52.127
-	Router – Core Link	192.168.52.128/30	255.255.255.252	.129 - .130	-	192.168.52.131
-	Reserved for Growth	192.168.52.132 - .255	-	-	-	-

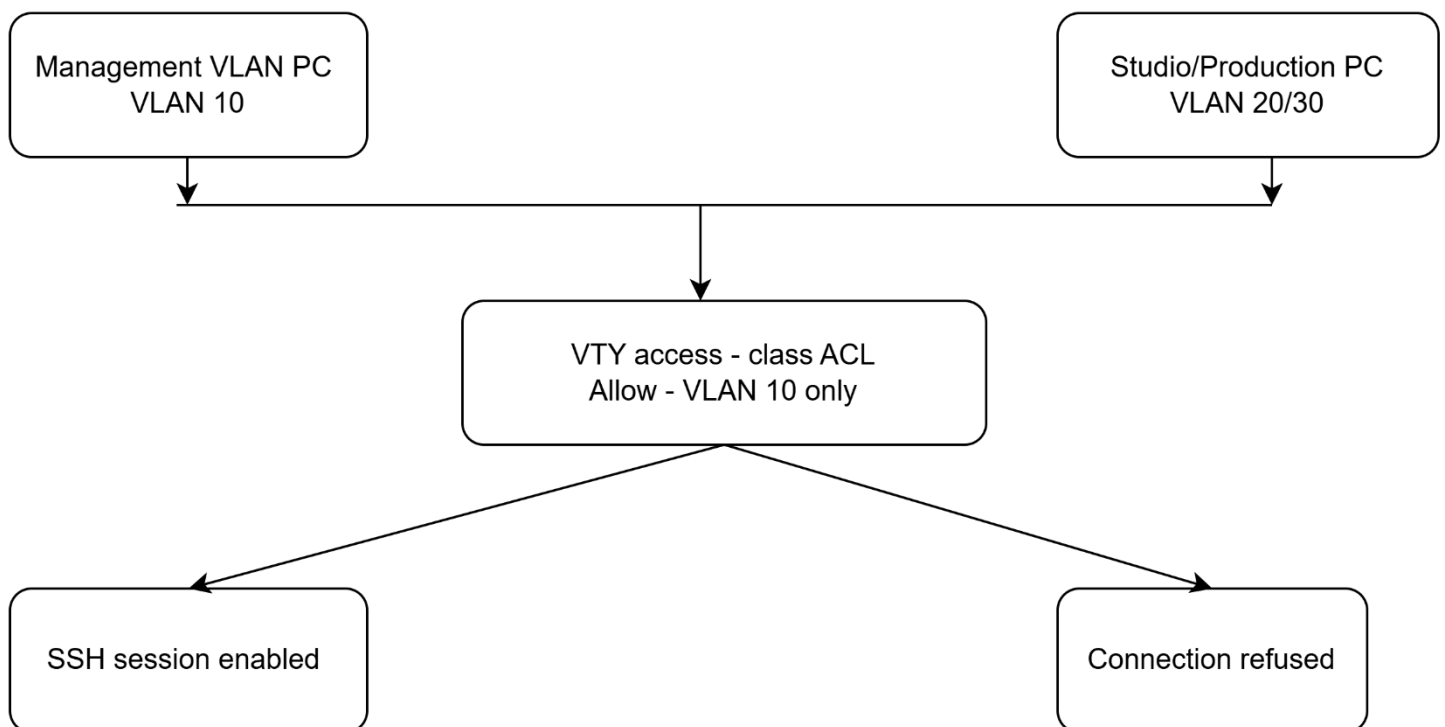
- VLSM was used instead of flat subnetting, so each VLAN's size roughly matches its expected device count, rather than wasting addresses.
- The /30 router - switch link uses the smallest possible subnet for a point-to-point link, which is standard practice and is efficient.
- Reserved space 192.168.52.132 - 255 is reserved for growth

7. CR14 – Guest Wireless Access Flow



8. Assigned Challenge (SSH) – Integration into the design

- SSH is configured on the router and core switch, the devices with VTY access; access-class ACL restricts SSH source addresses to VLAN 10 (Admin/Management) only. This means even a compromised device on Studio or Production VLANs cannot attempt management access.
- Telnet explicitly disabled, transport input ssh only. A specific, gradable configuration decision.



9. Github Repository

[wantedProgrammer/Computer-Networks](#)